Project Management

by PUL079BEI008
by PUL079BEI014

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Notes

- Our subject code is CT 658.
- This document encorporates for both, electronics and computer department..
- If a question was asked in Regular Exam, it is kept in **boldened** form. If it was asked in back exam, it is in regular font.
- Months are marked as:
  - Ba: Baisakh
  - Jth: Jestha
  - Asa: Ashar
  - Shr: Shrawan
  - Bh: Bhadra
  - Ash: Ashwin
  - Ka: Kartik
  - Mng: Mangsir
  - Po: Poush
  - Ma: Magh
  - Ch: Chaitra

Contents

1 Introduction

(2 Hours / 4 Marks)

1.1 Defintion of project and project management

1. Define project. [1] (**69 Ch**, 79 Ash, 74 Ash) [2] (**79 Ch**,**72 Ch**, 70 Asa)

2. How is project different from the operational work? [2] (**74 Ch**, 82 Ka, 75 Ash)
   |→Differentiate between project and operation. [2+2] (79 Ch)
   |→How is project different from day-to-day jobs? [1] (**69 Ch**)

3. Define project management. [1.5] (**81 Ch**)
   |→describe project management. [2] (**71 Ch**)

4. Why it project management important. [2] (**81 Ch**)

5. (Assumed) How can you relate the characteristics of project with your minor project? [4] (78 Ch)

6. Short notes on: Characteristics of project. [3] (80 Ash)
   |→list characteristics. [2] (**70 Ch**, 70 Asa)
   |→what are they? [1] (**69 Ch**)

1.2 Project Objectives

1. What are the triple constraints of project? [3] (**74 Ch**,**72 Ch**, 82 Ka, 75 Ash, 72 Ash)
   |→with figure, describe their relationship. [4] (74 Ash, 72 Ka)

2. Short notes on: Project tripple constaints [3] (**81 Ch**) [4] (**79 Ch**)

1.3 Classification of Projects

1. List out different bases of project classification. [5] (79 Ash)

2. How can IT projects be classified? [2] (73 Shr)

1.4 Project Life Cycle

1. How do you identify the completion of project phases? [1.5] (**81 Ch**)

2. What do you mean by project development life cycle? Explain various phases of PDLC. [5] (**74 Ch**)

3. What are the phases in project life cycle? [1] (70 Asa)

4. Explain the characteristics of project life cycle with diagram. [3] (74 Ash, 73 Shr)
   |→... of generic life cycle ... . [4] (72 Ka)

5. What are the phases in project life cycle? [2] (80 Ash)

2 Project Management Body of Knowledge

(4 Hours / 7 Marks)

1. What is PMBOK. [1] (73 Shr, 72 Ka) [2] (**79 Ch**,**74 Ch**,**69 Ch**, 82 Ka, 75 Ash, 70 Asa)
2. What are the knowledge contents that fall under PMBOK? [4] (73 Shr, 72 Ka)
   |→Describe what bodies of knowledge are required by a PM to contribute for a successful project implementation. [4] (**72 Ch**)
3. Write short notes on: Stakeholders. [2] (70 Asa)

2.1 Understanding of project environment

1. What do you mean by Internal, External and Task/operation environment in project? [2+2+2] (79 Ash)
2. How the external environmental influences on ICT project? [3] (**79 Ch, 78 Ch**,**74 Ch**, 82 Ka, 75 Ash)

2.2 General management skill, effective and ineffective project managers

1. What is the role of project manager? [1] (**70 Ch**)
2. What do you mean by general management skill? [2] (**81 Ch**)
3. One of the famous actors, Bill Cosby, once said "I don't know the key to success but the key to failure is trying to please everybody." How relevant is this statement in the field of IT System Project Management? What are other traits of successful Project Manager? Explain. [6] (**80 Ch**)
4. Explain role and responsibilities of key project members. [2] (**71 Ch**) [4] (74 Ash) [6] (**78 Ch**)
5. Differentiate effective and ineffective project managers, at least with 5 different characteristics. [6] (81 Ash)

2.3 Management Knowledge

2.3.1 Essential interpersonal and managerial skills

1. To be successful manager, what are the
   |→interpersonal & managerial skills [2+3] (**74 Ch**,**72 Ch**, 82 Ka, 80 Ash) [3+2] (75 Ash)
   |→managerial [2] (74 Ash)
   |→soft skills [3] (74 Ash)
   |→general management skills [5] (**69 Ch**)
2. What are suggested skills for project managers and for IT project managers? [4] (**70 Ch**)
   |→What are the different skill set required by a project manager? Briefly explain each of them. [5] (70 Asa)
3. Write short notes on: Expert Judgement. [3] (**69 Ch**)

2.3.2 Energized and initiator

2.3.3 Communication, influencing, leadership, motivator

2.3.4 Negotiation, problem solver, perspective nature, result oriented, global illiteracies

2.3.5 problem solving using problem trees.

1. Write short notes on: Decision Tree Analysis. [4] (**71 Ch**,**70 Ch**)

3 Portfolio and Project Management Institutes' Framework

(2 Hours / 4 Marks)

1. What is PMI? [1] (**70 Ch**)
2. How is PMI related to project management? [1] (**70 Ch**)
3. What is PMI Framework. [2] (75 Ash) [4] (79 Ash)
   |→Discuss PMI Framework in relation to project management. [2] (**72 Ch**)
   |→Discuss project management framework as per PMI along with the concept. [4] (**69 Ch**)
   |→Explain about PMI Framework. [4] (72 Ka, 70 Asa)
4. Explain knowledge areas of PMI framework. [4] (**71 Ch**)
5. "The PMI Framework provides a basic structure for understanding project management." Justify the statement. [5] (**74 Ch**)

3.1 Portfolio

1. Define project portfolio management. [2] (**79 Ch**, 81 Ash)
2. Compare project management with project portfolio management. [1] (72 Ka)
3. What are they key benefits of project portfolio management? [2] (74 Ash, 73 Shr) [3] (75 Ash)
4. Short notes on: Portfolio Management. [3] (**81 Ch**)

3.2 Project Management Office

1. Define project management office. [2] (80 Ash)
2. (Assumed) What is "project management practice"? [1] (**69 Ch**)

3.3 Drivers & Inhibitors of Project Success

1. Explain with example the concept of drivers of project success and inhibitors of project success. [2+3] (**70 Ch**)
2. Discuss briefly the drivers for making project success. [4] (81 Ash)
3. What are the reasons for success of ICT projects? [3] (80 Ash)
4. What are the major causes of failure of the ICT project? [2] (74 Ash) [3] (**72 Ch**)
5. Write about the challenges of IT projects. [2] (73 Shr)

4 Project Management

(4 Hours / 7 Marks)

4.1 Advantages of project management

4.2 Project management context as per PMI

1. Compare Project Manager versus Program Manager as per PMI. [4] (**80 Ch**)
2. Define PM context as per PMI. [1] (74 Ash)
3. Describe briefly about nine knowledge area under PMI. [6] (80 Ash)
4. Describe project management. [2] (**71 Ch**)

4.3 Management

4.3.1 Characteristics of project life cycles

4.3.2 representative project life cycles

4.3.3 IT Product Development Life Cycle

1. What is an SDLC? [2] (**69 Ch**)
2. Explain any one of SDLC you prefer in developing an IT project, and why. [5] (**69 Ch**)

4.3.4 Product Life Cycle and Project Life Cycle

1. Differentiate between product life cycle & Project life Cycle?
   [2] (**81 Ch**, 82 Ka) [3] (80 Ash) [4] (70 Asa)
2. What are the typical stages of Project Life Cycle, as per PMI? Write a brief note. [4] (81 Ash)

5 Project and Organization structure

(2 Hours / 4 Marks)

5.1 System view of project management

1. What is system view of project management? [2] (**81 Ch, 80 Ch, 79 Ch, 78 Ch**, 75 Ash, 74 Ash)
2. Explain about three sphere model of system management.
   [3] (**79 Ch**, 75 Ash, 74 Ash) [4] (**81 Ch, 80 Ch**)

5.2 Functional Organization, matrix organization

1. Define organization structure. [1] (75 Ash)
2. Most of the project follows functional organizational structure. If you agree, justify. [4] (**69 Ch**)
3. Describe types of organizational structure. [4] (**81 Ch**, 82 Ka) [6] (80 Ash)
4. Differentiate and compare among various organizational structures, with examples. [6] (**80 Ch**)
5. Describe about functional organization structure. [3] (**79 Ch**)
6. What are the different organizational structures, and which type of structure do you feel is the most effective in ICT project? [4] (75 Ash)
7. What are the disadvantages of projectized organizational structure? [2] (**79 Ch**)
8. Explain about Matrix organization
   |→and its type. [6] (79 Ash)
   |→with adv and disadv. [4] (70 Asa)
9. Explain various types of matrix organization. [4] (72 Ka)

5.3 Organizational structure influences on projects

1. Explain, how organizational structures influence projects? [3] (74 Ash)
   |→... IT projects. [3] (**78 Ch**)

6 Project Management Process Groups

(2 Hours / 4 Marks)

1. Discuss the concept of PM Process Groups (PGs). How is it related to project management knowledge area? [4] (70 Asa) [3+2] (75 Ash)
   |→ Give example of two processes with necessary inputs, tools and techniques and outputs. [4] (**70 Ch**)

6.1 Project Management process

1. (Assumed) Write short notes on Role of Feasibility Study [3] (**81 Ch**)
2. Describe about project process groups. (**79 Ch**)
3. Explain the concept of project management processes and discuss how process groups overlap within a project. [5] (81 Ash)

6.2 Overlaps of process groups in a phase, mapping of project management process groups to area of knowledge

1. Explain overlaps of project process groups with clear diagram. [5] (78 Ch)

7 Project Integration Management

(4 Hours / 7 Marks)

1. What is Project Integration Management? [1] (72 Ka) [2] (**79 Ch**, 75 Ash, 74 Ash) [3] (**78 Ch**)

7.1 Develop project charters

1. What is project charter? [1] (**81 Ch**,**71 Ch**, 80 Ash,)
2. Write short notes on: Project Charter. [3] (82 Ka)
3. What type of information should be included while making the project charter? [3] (79 Ash, 74 Ash)
   |→Explain with example. [3] (**79 Ch**) [4] (**81 Ch**)
4. How do you develop a project charter, explain the inputs and tools and techniques to develop it. [5] (**71 Ch**)
5. What are the essential information required to create Project Charter? [3] (**79 Ch**, 75 Ash, 74 Ash)
6. Explain the necessary inputs, tools and techniques and outputs to develop a project charter. [4] (72 Ka)

7.2 Develop preliminary project scope statement

7.3 Develop project management plan

7.4 Direct and manage project execution

7.5 Monitor and control project work

7.6 Integrated change control, close project, project scope management

1. Define WBS technique in scope management. [3] (**70 Ch**)
2. Discuss the process of defining project scope in more detail as a project progresses, going from infomration in a project charter to a project scope statement, WBS and WBS dictionary. [5] (**69 Ch**)
3. Explian about input, tools and techniques and output of integrated change control. [3] (73 Shr)
4. Explain about the integrated change control in detail. [4] (**70 Ch**) [5] (70 Asa) [6] (**72 Ch**)
5. What are the key features in change control on IT projects? [3] (73 Shr)
6. Short notes on: Statement of Work (SOW) [3] (**81 Ch**)
7. Imagine you have got the role of Project Manager for a software project that will make personalized and secure access of your semester-end mark-sheet possible. This will relieve Exam Control Office for printing semester end marksheet rather students will browse and download semester end mark-sheet as pdf only through either using browser or ECD app. (80 Ch)
   - Prepare a brief project scope document for the above system project work. [7]
   - While making the process groups, what are the important considerations that you have to manage and take care of? [5]

- (Assumed) The clothing manufacturer, Nepali Luga, is considering introduction a line of cargo pants made entirely from hemp. The proejct costs Rs. 4.6 million and will generate cash flows of NRs. 1 million for 5 years. What is the payback period? If the interest rate is 0.3% per month, what is the project's NPV? should the project be accepted? why or why not? [10] (**74 Ch**)
- Write short ntoes on: Break-even Point. [2.5] (**74 Ch**)

8. Describe about project scope management process. [4] (80 Ash)
   |→Define. [2] (73 Shr)
9. What are the essential components of project scope management? Explain. [5] (**71 Ch**)
10. Define project scope with example. [2] (79 Ash)
11. Compare Project scope with product scope. [2] (73 Shr)

7.7 Create Work Breakdown Structure

1. What is work breakdown structure? [1] (82 Ka) [1] (**70 Ch**)
2. Write importance of WBS. [2] (**70 Ch**)
3. Short notes on: WBS [3] (**81 Ch**)
4. What are the different ways/approaches to prepare a work breakdown structure for a project? [3] (**70 Ch**)
5. How does WBS affect the work estimate of the project duration/activity? Explain with example. [4] (82 Ka)
6. Discuss the process of defining a project scope in more detail as a project progresses, going from information in a project charter to a project scope statement, WBS and WBS dictionary. [5] (**69 Ch**)
7. Write short notes on: Responsibility assignment matrix. [3] (**72 Ch**)

7.8 Scope Verification

7.9 Score Control

8 Project Time Management

(4 Hours / 7 Marks)

8.1 Activity definition, decomposition of activities, activity attributes

8.2 Activity sequencing, precedence relationship

1. Why is it important to determine activity sequencing on projects? What are different diagrams/methods that cna be used to sequence activities in the project? [5] (70 Asa)

2. Why is there necessity of project time management? Explain how that is performed. [4] (**69 Ch**)

8.3 Network diagram, precedence diagram method, arrow diagramming method

1. Write short notes on: Arrow diagramming method. [3] (**69 Ch**)

2. Consider the following project and answer the followings. [2+2+2+1+2] (`82 Ka`)

Activity	Predecessor	Duration (week)
A	-	2
B	-	6
C	-	4
D	A	3
E	C	5
F	A	4
G	B, D, E	2
Total =		26 weeks

   - Draw the network diagram.
   - Draw the forward and backward pass.
   - Calculate the critical path with their respective activity.
   - Calculate the total duration of the project.
   - Find the total float of activity D, F.

3. Consider the following project and answer the followings. [2+2+1+2+2] (**`81 Ch`**)
   |→Same, but C has 5 days. [2.5x4] (**74 Ch**)

Activity	Predecessor	Duration (days)
A	-	11
B	-	15
C	A	8
D	A, B	10
E	B	5
F	C, D	4
G	E, F	7
Total =		60 days

   - Draw the network diagram.
   - Draw the forward and backward pass.
   - Calculate the total duration of the project.
   - Find the total float on each activity.

4. Consider the following project and answer the followings. [10] (**80 Ch**)

Activity	Predecessor	Duration (weeks)
A	-	2
B	A	3
C	A	5
D	B	1
E	A, B	2
F	C, D	4
G	E, F	3

- Draw the activity network diagram.
- Draw the forward and backward pass.
- Calculate the total float and free float.
- Identify the critical activity and critical path.

5. Consider the following project and answer the followings.

Activity	Predecessor	Duration (weeks)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D, E	4
H	F, G	3

10 (81 Ash)

- Draw the network diagram.
- Find the critical path.
- Calculate the late start and late finish.
- Calculate the total float time of activity E, F.

4+2+3+3 (79 Ash)

- Prepare network diagram
- identify critical activity.
- find project duration.
- find EST, EFT, LST, LFT, TF, FF, IF and independent float.

6. Consider the following activity table: [3+3+2] (80 Ash)

Activity	Predecessor	Duration (days)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D, E	4

- Draw the network diagram.
- Find the critical path and activity.
- Calculate the project duration.

7. Consider the following activity table: [1+2+2+1+2] (80 Ash)

Activity	Predecessor	Duration (days)
START	-	0
A	START	3
B	START	5
C	A, B	7
D	C	2
E	B	4
END	D, E	0

- Draw the network diagram.
- Calculate the forward pass and backward pass.
- Calculate the total float on each activity.
- Explain what critical path is.

8. Consider the following activity table: [2+2+2] (74 Ash)

Activity	Predecessor	Duration (days)
START	-	0
A	START	5
B	A	6
C	START	4
D	C	5
E	B	7
F	E	3
END	D, F	0

- Draw the network diagram.
- Calculate the critical path.
- Calculate the forward pass and backward pass and find total float time.

9. Produce a critical path network diagram showing the earliest start time and latest start times for each task, using the data. Calculate the total project time and total networkk slack time. [5+3+2] (73 Shr)

Tast Code	Task Name	Duration	Starts after completion of task(s)
PLAN	Plan project	3	
REQ	Capture Requirements	8	PLAN
AGREE	Agree requirements with customer	2	REQ
DESIGN	Design system	10	AGREE
CODE	Code System	12	DESIGN
ID	Identify subcontractors	3	DESIGN
BUY	Buy-in subcontractor code	5	ID
INTEG	Integrate code and buy-in code	6	CODE, BUY
TRAIN	Train staff	5	DESIGN
REL	Release system	4	INTEG, TRAIN

10. Consider teh below table as the different WBS related job and sequences as per the project plan for MIS building project. Times listed in weeks and the activity network proceeds from 1^{st} node to 10^{th} node following the table sequences. Draw the critical path and network diagram and calculate the total project time and total network slack time of this project. [5+3+3] (73 Shr)

Job	Initial node	Final node	Estimated Time
A	1	2	3
B	1	3	2
C	1	4	3
D	2	5	3
E	2	9	2
F	3	5	2
G	3	6	2
H	3	7	3
I	4	7	6
J	4	8	2
K	5	6	2
L	6	9	3
M	7	9	5
N	8	9	3
O	9	10	2

11. A big software project is under consideration for development. Overall 10 different activities as WBS are identified as listed below table with their timing in number of weeks. [3+3+3] (72 Ka)

Activities	a	m	b
(1,2)	7	11	13
(2,3)	1	4	6
(2,4)	10	15	48
(3,5)	12	20	26
(3,6)	4	7	16
(3,7)	4	7	16
(6,7)	5	8	11
(4,7)	2	8	14
(7,8)	9	12	15
(8,9)	1	4	7

where a = optimistic, b = pessimistic, m = most likey

Calculate the following:

a. What is the expcted time of completion of the project?

b. What is the probability of completing the project in 34 weeks?

c. What is the probability of activity 7 being completed in the twentififth week?

12. A project work consists of the following activities: [9] (**71 Ch**)

Activity	Description	Duration (in days)
A(1-2)	Start earth work	3
B(1-4)	Vendor Selection	2
C(1-7)	Start handling	1
D(2-3)	Continue earth work	3
E(3,6)	Finish earth work	2
F(4-5)	Ordering raw material	4
G(4-8)	Excavation for drains	6
H(5-6)	Receiving raw material	5
I(6-9)	Base concreting	4
J(7-8)	Continue handing	4
K(8-9)	Laying drains	5

Draw the network diagram and trace the critical path of the network. What are the various timings and the total duration of the above project?

8.4 Activity resources estimating

8.5 Determining resource requirements

1. (Assumed) Compare: Consistency vs completeness in requirements engineering. [3] (**72 Ch**)

8.6 Schedule development and control

8.7 Principles of scheduling

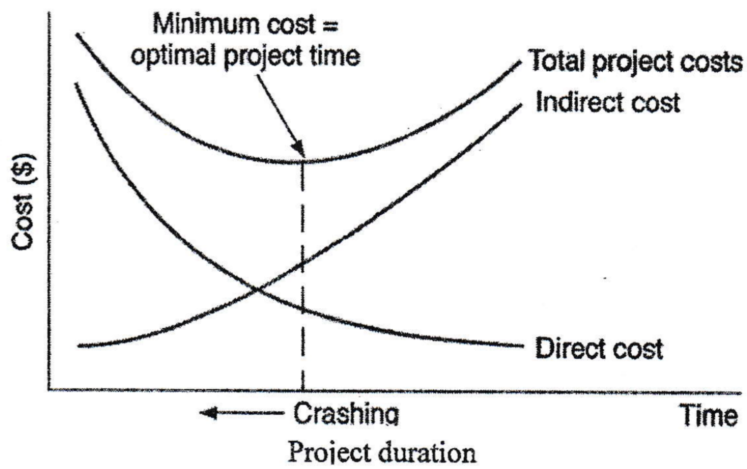
8.7.1 milestones, forward pass, backward pass

8.7.2 Critical path method, critical chain technique

1. Write short notes on:
   |→Critical path analysis. [4] (**70 Ch**)
   |→Critical Chain scheduling. [2] (70 Asa)

8.7.3 Gantt chart, schedule control

1. Project Crashing is termed as optimal point of trade-off between project cost and project time for any project while assessing delayed or over-run project. Do you agree or not? In any case justify with the help of right-side graph. [6] (**80 Ch**)



9 Project Cost Management

(4 Hours / 7 Marks)

9.1 Cost and project, cost management

9.2 Cost estimating, types of cost estimates, estimating process and accuracy

1. What is cost estimating? [1] (**69 Ch**)
2. Explain different tols and techniques used for cost estimating. [4] (**69 Ch**)
3. Write short notes on: COCOMO for ICT project. [2] (70 Asa) [2.5] (**74 Ch**) [3] (82 Ka, 72 Ka)

9.3 Cost estimating tools

9.4 Cost budgeting, cost aggregation, deriving budget from activity cost

9.5 Cost control process, cost control methods, earned value management

1. Compare Project Cost management versus Earned Value management. [3] (**80 Ch**)

9.6 EVM benefits, variance analysis

1. What is EVM? [1] (**79 Ch**,**74 Ch**, 80 Ash, 74 Ash, 72 Ka) [2] (**81 Ch, 78 Ch**, 82 Ka)
   |→What is EV analysis? [2] (73 Shr)
2. Derive the formluas that are used in schedule and cost performance, explain their signficances. [3+3] (72 Ka)

9.7 Numerical

1. Suppose you have project with total budget Rs. 500,000/- scheduled to last for 8 months, you check your records and find that you have spent Rs. 300,000/- so far. The team has completed only 40% of the project work but when you check the schedule, it says they should have to complete 50% of the work. Now calculate EV, PV, SPL, CPI, EAC, and find the new estimate duration for above project. [1+1+1+1+1+1] (82 Ka)
2. Your project is scheduled for 2 years. Nine months into the project, while the total project budget is Rs 42,00,000. You have already spent Rs 16,50,000. CPI is 0.875. Calculate EV, EAC, ETC, VAC and is the project is behind the schedule or ahead schedule? [1+1+1+1+2] (**81 Ch**)
3. Suppose you are the project manager on an IOT project. The project is set to be completed in 10 months with an estimated cost of Rs 500,000. The project has been running for 5 months now, the team has spent Rs 220,000 and completed an amount of work worth Rs 255,000. Calculate CPI and SPI and Share your conclusion. [1+2+2+1] (79 Ch)
4. Calculate the schedule and cost variance for a project whose estimated budget is $100,000 and estimated time is 50 man months. 5 people were involved in the project. After 7 months, 60% of work was completed and $80,000 was expended. Also estimate the new cost and time for the project. [5] (**78 Ch**)

5. The project was scheduled to be completed in 12 months and budget as Rs 1,20,000. After 4 months, it was found that Rs. 50,000 has been spent and 40% of the work has been completed. [2+2+2] (82 Ash)

    - Calculate the cost variance and schedule vriance for the project.
    - Calculate the cost performance index and schedule performance index.
    - Calculate the estimated cost and time for completing the project if the project continues without any changes.

6. Your project is scheduled for 2 years. There are six different teams working on five major functional areas. Some teams are ahead of schedule while others are falling behind. There are more cost overruns in some areas but you have also saved costs in others. Due to all this, it is difficult to understand whether you are over or under budget. Nine months into the project, while the total project budget is Rs 42,00,000/- you have already spent 16,50,000/- CPI is 0.875/-. Can you perform EV analysis and budget forecast? and find the value of EV and VAC. Is the project over or under bduget? [2+1+2+1] (80 Ash)

7. Suppose you have a project that is scheduled to be completed in 10 days aat a budgeted cost of 100000/- at the end of day 6, you do analysis and you determine the job is 70% complete, and you have spent 65,000/-. (79 Ash)

    - What is the project's EV at Day 6? [1]
    - What is the project's budget at completion? [1]
    - Is the project ahead, behind schedule or on time? [2]
    - Is the project expected to complete on budget, under or over budget? [2]
    - What is the project's SPI nd CPI at Day 6. [2]

8. Suppose you have IT project which might look after project planning:

Task ID	Name	Start	End	Budget
101	Setup Database	Step 1	Sept 10	Rs. 1,00,000 /-
102	Build Application	Step 7	Sept 20	Rs. 1,50,000 /-

    Lets assume project has started on Sep 3^{rd} determine that the first task is 20% complete and second task is 10% complete and budget at completion is Rs 2,50,000 and after reviewing your time and expenses software and compiling any miscellaneous expenses, we determine that the actual cost of the first task is Rs. 45,000 and second task of actual cost Rs 20,000. Find the SV and CV of the project. Is the project over budget or under budget? [2.5+2.5+1] (75 Ash)

9. Suppose you are three month into six month project. Assume that the budget burn rate is constant and BAC is Rs. 1,20,000 and AC is 65,000, SPI is 1.2. Find the CPI of the project and EAC. Is the project over budget or under budget? [2+1+1] (**74 Ch**)

10. The project has been planned that total estimated cost of project is Rs 5 Lakhs and there are 20 widgets to complete in 10 months duration. at 5^{th} months, it reported that project was completed 40% and it has been spent Rs. 3 lakhs. Now calculate CV and SV. Is the project behind or ahead of schedule? [2+2+1] (74 Ash)

11. A project is scheduled for the time of 12 months. The estimated cost of project is \$400000. After 3 months, the evaluation is done, and it it identified that 40% of work is accomplished but \$200000 cost has been incurred. Now calculate cost and schedule variance of the project. [6] (73 Shr)

12. Suppose you are managing a software development project. The project is expected to be completed in 8 months at a cost of Rs. 50,000 per month. After 2 months, you realize that the project is 30 percent completed at a cost of Rs. 200,000. Determine whether the project is on-time and on-budget after 2 months. Calculate Cost and Schedule performance index. [8] (**72 Ch**)

13. If SPI is 0.75 in a mega project undergoing near Devikapur district with earned value of being 60. Now calculate the planned value and also state whether the project is ahead schedule or behind schedule. [6] (**70 Ch**)

14. Given the following information, for 1 year project, use EVM method to calculate, CV, SV, CPI and SPI for the project. [6] (70 Asa)
    Planned Value = NRs 23,000
    Earned Value = NRs 20,000
    Actual Cost = NRs 25,000
    Budget at Completion = NRs 1,20,000

15. If earned value is twice its actual cost for a project, calculate its cost performance index and cost variance percentage. Is the project over/under budget? [6] (**71 Ch**)

10 Project Quality Management

(3 Hours / 5 Marks)

10.1 Quality theories

1. How to improve quality in ICT project. [2] (82 Ka)
2. Define project quality. [2] (80 Ash)
3. What do you understand by quality in context of project management? [1] (**69 Ch**)
4. Is there always a tradeoff between quality and productivity? Explain with IT related example. [6] (**71 Ch**)

10.2 Quality planning, project quality requirements, cost of quality, quality management plan

10.3 Quality assurance, quality audit, approach to quality audit

1. Quality is one the most important factors to be considered for effective delivery of project objectives. How quality assurance and quality control is implemented in order to deliver a successful project? [2] (**81 Ch**) [3] (**79 Ch**) [5] (**79 Ch**, 74 Ash) [8] (**72 Ch**)
2. Write short notes on: Quality Audit Plan. [3] (72 Ka)
3. How is quality assured in software industry? [2] (79 Ash)
4. Compare: Traceability vs Adaptability in reviewing steps. [3] (73 Shr)

10.4 Quality control process, control chart, pareto charts, testing of IT system, the test life cycle

1. What are the possible steps to imporve project quality. [4] (**71 Ch**)
2. What is quality planning, quality assurance and quality control. Explain different approaches to these projects. [4] (**70 Ch**)
3. Discuss the quality control process and its major outputs. [4] (**69 Ch**)
4. How the control charts used to monitor the quality of project result? [3] (82 Ka)
5. Comare: control charts vs Cause and effect charts for quality assessment. [3] (73 Shr)
6. Write short notes on: TQM. [3] (**81 Ch**)
7. What are tools and techniques for TQM? [4] (73 Shr)
8. DeMacro states "you cannot control what you cannot measure". Considering from software project manager's perspective, justify with relevant example. [] (**72 Ch**)
   The question is not even indexed, its just there, with no marking assigned. not even the dignity to say this question is no. 4 or anything.

9. A project manager can modify three basic elements of software proejct: the resources available, the time available and the amount of product to be built. Describe how each of these three can be varied during a development process in order to ensure the resulting software is of high quality. [6] (**72 Ch**)

10. Write steps to improve quality IT project. [3] (73 Shr)

11. Explain 80/20 rule of the pareto principle with exmaple. [3] (80 Ash)

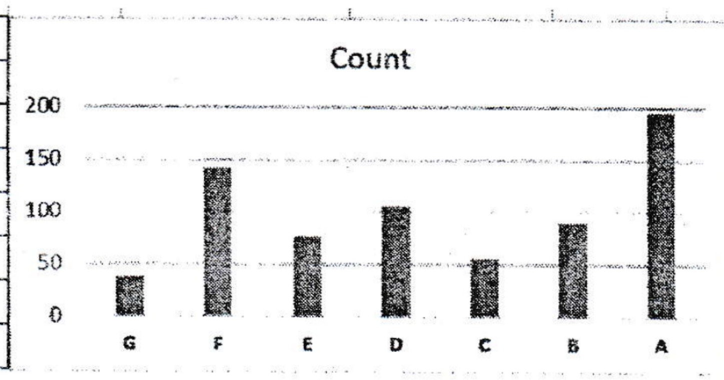
12. How pareto charts help to achieve better quality project? [5] (**74 Ch**)

13. Write short notes on: Pareto Analysis. [3] (72 Ka)

14. Quality is one the most important factors to be considered for effective delivery of project objectives. How quality assurance and quality control is implemented in order to deliver a successful project? [2+3] (**81 Ch**)

15. Consider below data about customer complaint on e-commerce portal and their srevices. Draw a free-hand sketch of Pareto diagram after having pareto analysis with clearly showing all necessary calculations. As per Pareto principles, which are the first three complaints should be addressed first? What will be the coverage of issues by these solving such three issues? [6+2] (**80 Ch**)

Code	Complaint Types	Count
G	Unexpected fees	37
F	Unresponsive customer support	142
E	Lack of security	76
D	Poor display on mobile	105
C	Wrong product delivery	55
B	Product different than display	90
A	Slow website speed	195



16. ISO/IEC 5055:2021 is an ISO standard for measuring the internal structure of a software product on four business-critical factors: Security, Reliability, Performance Efficiency and Maintainability. If you are building a typical software and associated support system of accounting and financial audit record management system for academic Institutes and universities, how you will be using ISO/IEC 5055:2021 for ensuring quality of your system? [8] (81 Ash)

17. Differentiate quality assurance and quality control. [2] (79 Ash)

18. How quality assurance and quality control are implemented in order to deliver a successful project? [5] (75 Ash)

19. Write short notes on: Cause and Effect Diagram. [2.5] (**74 Ch**)

11 Project Communication Management

(3 Hours / 5 Marks)

11.1 Importance of communication management

1. What is project communication management? [1] (**81 Ch**)
2. Why communication management is important in IT project? [2] (**79 Ch, 78 Ch**) [3] (80 Ash, 74 Ash)
3. Why better communication management is critical for projects? Discuss communication management plan that should be considered for ICT Projects. [5] (**70 Ch**,**69 Ch**)
4. Why it is important to complete project successfully? [2] (**81 Ch**)
5. What is difference between communication skill and communication management? [2] (**72 Ch**)

11.2 Communications planning process, communication requirement analysis, organizing and conducting effective meeting

1. Explain briefly on project communication planning. [3] (81 Ash)
2. Explain the hazards of communication error in a project. [2] (82 Ka, 79 Ash) [3] (75 Ash, 74 Ash)
3. Find the no. of communication channels among 7 people. [2] (**81 Ch**)
4. Compare Formal versus informal communication in PM. [3] (**80 Ch**)
5. Compare: Horizontal vs Vertical communication and their degree of formalness. [3] (73 Shr)
6. As a project manager how do you conduct meeting effectively? [2] (80 Ash, 74 Ash) [3] (**79 Ch**)
   |→What are the steps to conduct effective meeting? [2] (81 Ash)
7. As project manager, how do you write effective email? [2+2] (**78 Ch**)
8. How does communication skill help to resolve conflicts in ICT project? Explain with example. [4] (**72 Ch**)

11.3 Information distribution process

1. Explain about the necessity of information and distribution and its tools and techniques. [5] (70 Asa)

11.4 Performance reporting process, integrated reporting system

1. Explain various tools and techniques for performance reporting. [5] (**71 Ch**)
2. Why reporting system is required in project? [2] (79 Ash) [3] (82 Ka)
3. Why documentation and reporting system is required in a project? [2] (75 Ash, 74 Ash)
4. A project manager is having trouble getting a project membeer to complete their tasks as assigned. What type of communication would the manager want to use to address this problem initially and why? [5] (**74 Ch**)

12 Project Risk Management

(4 Hours / 7 Marks)

12.1 Understanding Risk, Project risk

1. What are the different sources of risk in IT project? [2+3] (79 Ch)
2. Explain different types of Risk and illustrate the Risk management model with block diagram according to PMI, Project Risk Management Process. [7] (72 Ka)
3. What is project risk? [1] (**78 Ch**)
4. Write short notes on: Categories of Risk. [2] (70 Asa)

12.2 Risk Management planning process, risk management plan

1. What is project risk management? [1] (**69 Ch**) [2] (**81 Ch, 79 Ch**)
2. Explain about risk management plan. [4] (**78 Ch**)
3. Discuss the processes involved in risk management plan briefly. [4] (**69 Ch**)
4. Why risk management is an essential part of project management? [2] (80 Ash)

12.3 Risk identification, risk identification techniques

1. What are the risk identification techniques? [3] (**81 Ch**)
2. Discuss briefly on risk identification techniques. [2] (**80 Ch**)
   |→in ICT project. [3] (80 Ash) [5] (74 Ash)
3. Write short note on: Delphi Technique. [2] (75 Ash)
4. Create a sample risk register of any ICT related project. [3] (**80 Ch**)

12.4 Qualitative risk analysis process

1. Short notes on: Quantitative Risk Analysis [3] (**81 Ch**)
2. Describe different methods of qualitative analysis of risk. [4] (79 Ash)
3. (Assumed) Short notes on SWOT Analysis. [3] (82 Ka, 80 Ash, 75 Ash, 73 Shr)
4. Write short notes on: Project management information system. [2] (70 Asa)

12.5 Quantitative risk analysis process, modeling techniques

1. Describe different methods of quantitative analysis of risk. [4] (79 Ash)

12.6 Risk response planning, resolution of risk

1. Write short notes on: Sensitivity Analysis. [4] (**71 Ch**)
2. Why risk response planning is important in project? [2] (**74 Ch**, 75 Ash)
3. Describe three typical risks that can occur in a software project and for each of these risks suggest two possible countermeasures. [8] (73 Shr)
4. Does effective communication management skill reduce the associated risk of an IT proejct? Explain with example. [4] (72 Ka)

12.7 Strategies for negative risks or threats, strategies for positive risks or opportunities

1. What are the response strategies for negative risk? [3] (**74 Ch**, 75 Ash)

12.8 Risk monitoring and control process

12.9 Sensitivity

1. Short notes on: Tornado Analysis. [3] (81 Ash) [4] (**70 Ch**)
2. Compare: Decision Tree vs Tornado Analysis for risk management. [3] (**72 Ch**)

13 Project Procurement Management

(3 Hours / 5 Marks)

13.1 Procurement management process flow

1. What are the benefits of procurement management? [2] (82 Ash)
2. What do you mean by project procurement management? [1] (75 Ash, 74 Ash) [2] (**81 Ch, 80 Ch**,**74 Ch**,**71 Ch**, 82 Ka, 80 Ash)
3. What are the different processes adopted for procurement. [3] (**71 Ch**)
4. Explain about procurement management process flow. [2+3] (**79 Ch**)
5. What is procurement process? How is it performed in a project? [1+4] (70 Asa)
6. What makes the procurement process of very crucial component in project management? [2] (72 Ka)

13.2 Plan purchases and acquisition process, enterprise environmental factor

1. Compare Quoatation based purchase vs Tender based purchase for procurement process. [3] (**72 Ch**)
2. Consider you are hired as a consultant in an IT college where every year 50 students are admitted in 4 year program. You are asked to prepare a tender. Specification document for setting up a digital library to be set-up on that college. State your all assumptions that you will be making while preparing the document. [6] (**72 Ch**)

13.3 Organizational process assets

1. Short notes on: Organizational Process Asset. [3] (82 Ash)

13.4 Plan contracting process, standard forms, evaluation criteria

13.5 Processes

1. Explain the different processes adopted in the procurement management?
   [3] (**80 Ch**) [4] (**81 Ch**,**74 Ch**,**74 Ash**, 82 Ka, 80 Ash, 75 Ash)
2. Discuss the process of procurement in any project. [5] (79 Ash)

13.5.1 Request seller response process

13.5.2 Select seller process

13.5.3 Contract Administration process

1. (Assumed) List out the components of bidding document. [3] (81 Ash)
2. What are typical issues to be considered in e-bidding as procurement processing tool? [4] (72 Ka)

13.5.4 Contract closure process

1. Write short notes on: Contract Closure Procedure. [2] (75 Ash) [3] (73 Shr, 72 Ka) [4] (74 Ash)

14 Developing Custom Processes for IT Projects

(3 Hours / 5 Marks)

14.1 Developing IT project management methodology

1. Discuss about IT project management methodology. [5] (70 Asa)
2. (Assumed) Discuss briefly the challenges in IT project in Nepal. [4] (**80 Ch**)
3. (Assumed) Write about the challenges of IT projects. [2] (73 Shr)
4. (Assumed) What are the reasons for failure of the ICT projects? [5] (**78 Ch**)
5. (Assumed) What is system development methodology? [2] (81 Ash)
6. (Assumed) Mention atleast three different methods and compare with Agile development methodology. [4] (81 Ash)
7. (Assumed) Being an IT project manager how are you going to manage an IT based project that demands regular updates with new trends in market. [5] (**70 Ch**)
8. Write short ntoe son: Six Sigma. [4] (**71 Ch**)

14.2 Moving forward with customized management process

1. Short notes on: Need custom processes for IT project [4] (**79 Ch**)[2] (**78 Ch**)
   |→Why do you need custom processes for IT Projects? [4] (80 Ash)
2. Small IT companies can't follow all the suggestions as per PMI, so they can develop their own custom processes for managing project. Do you agree? Discuss. [3] (79 Ash)
3. Write short notes on: Out sourcing and off-shoring options. [4] (**70 Ch**, 74 Ash)

14.3 Certified associate in project management

14.4 Project management maturity

1. What is a maturity model for software development? Explain them. [5] (70 Asa)
2. Write short notes on: Project Management Maturity. [4] (**71 Ch**)
3. Write short notes on: PMMM. [3] (**72 Ch**, 82 Ka)

14.5 Promoting project Excellency through awards and assessment

1. What are the roles of award and assessment in achieving excellency in project completion, briefly explain. [5] (**69 Ch**)

14.6 Certification process flow

14.7 Chode of ethics

1. Write short notes on: ICT Code of Ethics. [2] (75 Ash) [3] (82 Ka) [4] (74 Ash)

14.8 Future trends

1. Short notes on: Future trend of IT projects [4] (**79 Ch**)
2. Short notes on: Possible impact of AGI on Software project management. [3] (81 Ash)
3. Write short notes on Trends in cloud computing. [4] (**70 Ch**)

15 Balanced Scorecard and ICT Project Management

(1 Hours / 2 Marks)

1. Write short notes on: Balanced Scorecard.
   [2] (**78 Ch**, 70 Asa)
   [2.5] (**74 Ch**)
   [3] (**81 Ch**,**72 Ch**,**69 Ch**, 82 Ka, 81 Ash, 80 Ash, 75 Ash, 73 Shr, 72 Ka)
   [4] (**71 Ch**,**70 Ch**, 74 Ash)

2. Compare: KPI versus Balanced scorecard. [4] (**80 Ch**)

3. (Assumed) Short notes on: Problem in ICT Project. [3] (80 Ash)